

Network-Based Distance Between Drug and Disease Targets (Guney et al.). Guney. et al.[1] define the network-based distance between a drug and a disease as the average of shortest distances from drug target proteins to the closest disease target protein (d_{asc}) within a protein–protein interaction (PPI) network. Specifically, for each drug target protein, the shortest path to any disease target protein is computed, and these minimal distances are then averaged across all drug targets.

To illustrate this concept, Fig. 1 presents a simple example of a drug–disease association in a PPI network. The drug targets proteins s_1 and s_2 , while disease effects proteins t_1 and t_2 and t_3 . For each drug target protein, the shortest path to the disease target set $\{t_1, t_2, t_3\}$ is identified. In this example, the closest disease target to s_1 is t_2 , yielding a shortest path length of 2. Similarly, the shortest path from s_2 to the disease target set has length 3 targeting t_3 . The d_{asc} between the drug and disease target sets is therefore computed as $(2+3)/2=2.5$.

This example demonstrates how network proximity is quantified in the Guney et al.[1] framework and forms the basis for assessing whether a drug is closer to disease-associated proteins than expected by chance.

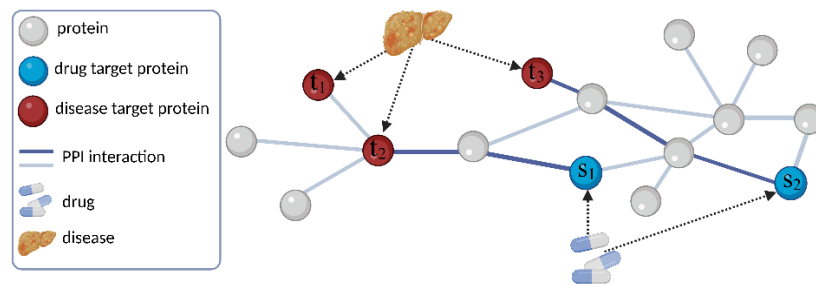


Fig. 1. Illustration of PPI networks and set of disease target proteins t_1 , t_2 and t_3 and drug target proteins s_1 and s_2 .